

Siraj T M

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Date of birth: 30/07/2003

PUBLICATIONS

- [2026]

Graph-Guided Change Detection and State Space Model Captioning with Semantic Uncertainty Estimation
Authors: Siraj T. M, Reji V., R. S. Mathew | **Journal Name:** MIGARS 2026 Conference
- [2025]

Context-qsmnet: Enhancing Generalization in Quantitative Susceptibility Mapping with Context-Aware Deep Neural Networks
Authors: Thomas Apotheker, Siraj TM, Raji Susan Mathew, N. Awasthi | **Journal Name:** ICVGIP 2025 Conference

EDUCATION AND TRAINING

BSMS - Data Science Major
IISER Thiruvananthapuram iisertvm.ac.in
Address: IISER Thiruvananthapuram, Maruthamala, Vithura, 69551, Thiruvananthapuram, India
Total CGPA : 8.00/10
Major CGPA : 8.40/10

WORK EXPERIENCE

- Dr.Raji Susan Mathew*
City: Thiruvananthapuram | **Country:** India

[01/08/2025 – 29/05/2026]

Master's Thesis

M1: Developed an interpretable voxel-wise deep reinforcement learning framework for quantitative susceptibility mapping reconstruction (RL-QSM)
M2: Extended RL QSM by incorporating model-based unrolling into the existing framework.

Dr.Raji Susan Mathew and Dr. Reji J
City: Thiruvananthapuram | **Country:** India
- [01/11/2024 – 07/07/2025]

Internship

Developed graph-guided change detection and state space model captioning for remote sensing images with semantic uncertainty estimation

Dr.Raji Susan Mathew and Dr. Navchetan Awasthi
City: Thiruvananthapuram | **Country:** India
- [01/12/2024 – 01/05/2025]

Internship

Contributed to the development of a generalizable model for QSM reconstruction using contextual input-target pairs.

Dr. Biplab Banerjee
City: Mumbai | **Country:** India
- [01/06/2024 – 30/08/2024]

Summer Intern

Worked on remote sensing change detection using multimodal vision-language representation learning using the CLIP model

HONOURS AND AWARDS

- [01/03/2022]

Innovation in Science Pursuit for Inspired Research (INSPIRE) Awarding institution: DST-GOI

INSPIRE Scholarship awarded by DST-GOI Awarded to the top 1% of students based on Class 12 board exam performance. Scholarship for pursuing Natural and Basic Sciences at the BSMS level.

SKILLS

Programming Languages

Python | MATLAB | C | Bash | R | C++ | LaTeX | MySQL

Frameworks & Packages

Pytorch | OpenCV | MPI | Git | OpenMP | Tensorflow | CUDA | NetworkX | Linux

Machine Learning

Deep Learning | Computer Vision | Reinforcement Learning | Graph Neural Networks | Vision
-Language Models